

OUARZAZATE, Morocco

WMO: 602650

Lat: 30.939N

Lon: 6.909W

Elev: 1153

StdP: 88.22

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.2	(c) 1.9	(d) -17.1	(e) 1.0	(f) 14.5	(g) -14.8	(h) 1.2	(i) 15.8	(j) 15.1	(k) 13.8	(l) 13.2	(m) 12.2	(n) 1.2	(o) 300	(p) 0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 15.3	(c) 39.2	(d) 16.3	(e) 38.8	(f) 16.2	(g) 37.8	(h) 16.0	(i) 19.0	(j) 31.4	(k) 18.2	(l) 31.3	(m) 17.5	(n) 31.5	(o) 4.9	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.8	12.9	21.7	14.1	11.6	21.9	12.8	10.6	22.6	58.9	31.4	56.0	31.0	53.8	31.0	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.5	(b) 10.4	(c) 8.6	(e) -2.2	(f) 40.8	(g) 1.7	(h) 0.9	(i) -3.5	(j) 41.4	(k) -4.5	(l) 41.9	(m) -5.5	(n) 42.4	(o) -6.7	(p) 43.0		
(a) 12.5	(b) 10.4	(c) 8.6	(e) -5.1	(f) 20.5	(g) 1.8	(h) 1.5	(i) -6.3	(j) 21.6	(k) -7.3	(l) 22.5	(m) -8.3	(n) 23.4	(o) -9.6	(p) 24.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.0	9.8	12.1	15.7	19.5	23.3	27.6	30.8	29.8	25.5	20.6	14.7	10.4		
	DBStd	7.62	2.52	2.58	2.77	3.06	3.21	2.29	1.53	1.98	2.53	2.58	3.07	2.16		
	HDD10.0	69	34	10	1	0	0	0	0	0	0	0	3	21		
	HDD18.3	923	264	174	87	24	4	0	0	0	0	10	114	245		
	CDD10.0	3730	28	69	179	284	413	529	645	612	464	328	143	34		
	CDD18.3	1541	0	0	7	58	158	279	387	354	214	80	4	0		
	CDH23.3	21362	1	8	154	693	1968	3891	6103	5112	2549	807	76	0		
	CDH26.7	12052	0	0	21	194	913	2273	3979	3146	1296	224	6	0		
(14) Wind	WSAvg	2.8	2.1	2.5	3.2	3.7	3.7	3.2	3.2	3.0	2.8	2.3	2.2	1.7		
(15) Precipitation	PrecAvg	88	10	9	13	5	5	2	2	9	13	15	14	8		
	PrecMax	213	77	55	99	29	45	10	15	32	67	83	91	45		
	PrecMin	19	0	0	0	0	0	0	0	0	0	0	0	0		
	PrecStd	42	15	14	18	7	9	3	3	9	14	20	22	11		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.8	24.7	28.9	32.1	35.9	39.1	40.9	40.0	37.2	32.2	27.4	22.0		
		MCWB	9.2	9.6	11.3	13.0	14.4	15.6	16.7	16.3	16.4	14.1	12.6	9.5		
	2%	DB	20.7	22.8	26.8	30.2	34.4	37.9	39.9	39.0	35.7	30.8	25.4	20.1		
		MCWB	8.6	9.3	10.8	12.6	14.1	15.4	16.4	16.5	15.5	13.9	11.8	8.8		
	5%	DB	18.9	21.1	25.0	29.0	33.1	36.8	39.1	38.1	34.5	29.1	23.8	18.7		
		MCWB	8.0	8.7	10.3	12.2	14.0	15.3	16.2	16.3	15.1	13.6	11.3	8.5		
	10%	DB	17.0	19.5	23.2	27.2	31.8	35.4	38.2	37.0	33.1	27.8	22.0	16.9		
		MCWB	7.0	8.2	9.8	11.5	13.7	14.9	16.1	16.3	15.0	13.3	10.6	8.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	12.0	13.4	14.8	18.0	18.1	19.5	20.5	19.0	16.9	14.4	12.3		
		MCDB	19.8	18.9	21.8	27.5	29.4	31.5	35.2	31.8	28.1	25.2	23.6	17.4		
	2%	WB	9.5	10.8	12.2	13.7	16.2	17.2	18.5	19.2	17.9	15.8	13.3	10.7		
		MCDB	18.0	18.1	22.2	26.8	29.3	31.0	34.6	31.0	28.1	24.0	21.8	16.3		
	5%	WB	8.6	9.9	11.3	12.9	15.2	16.6	17.7	18.5	17.1	15.1	12.2	9.7		
		MCDB	17.1	18.1	21.9	26.2	28.6	30.9	35.0	30.7	28.4	23.7	21.0	15.8		
	10%	WB	7.7	9.0	10.5	12.2	14.4	16.0	17.0	17.8	16.4	14.4	11.3	8.8		
		MCDB	15.9	17.7	21.2	25.4	28.3	30.8	35.1	30.7	28.1	24.0	20.3	15.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.4	14.0	13.8	13.7	14.2	14.9	15.3	14.5	14.1	13.2	13.7	13.7		
		MCDBR	17.2	16.9	16.5	15.5	15.8	16.2	16.2	15.6	15.9	15.1	15.7	16.1		
		MCWBR	8.9	8.1	6.9	5.7	5.5	5.5	5.4	5.0	5.3	5.7	7.1	8.2		
		MCDBR	15.4	14.1	14.3	14.2	14.0	14.3	14.6	13.4	13.3	12.4	13.5	13.2		
	5% WB	MCWBR	8.2	6.8	6.2	5.7	5.6	4.9	5.2	4.8	4.7	5.1	6.4	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.263	0.305	0.368	0.388	0.429	0.475	0.562	0.531	0.435	0.387	0.316	0.282		
	taud		2.601	2.456	2.234	2.199	2.080	1.957	1.751	1.843	2.102	2.245	2.480	2.560		
	Ebn at Noon		986	961	912	904	864	820	749	768	834	853	906	942		
	Edh at Noon		79	101	135	145	165	186	228	205	152	123	89	79		
(44) All-Sky Solar Radiation	RadAvg		3.92	4.79	5.97	7.00	7.57	7.97	7.52	6.79	6.00	4.86	3.99	3.54		
	RadStd		0.23	0.34	0.32	0.33	0.43	0.44	0.34	0.28	0.38	0.26	0.27	0.22		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.50	N/A	-2.22	+0.62	-0.76	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	+0.48	N/A	-1.85	+0.67	-0.60	N/A	N/A	N/A	+153	+171

Nomenclature: See separate page